Quantum School of Technology

Roorkee - Dehradun Highway, Mandawar, Roorkee, Uttarakhand 247167

☎ **Fax No.**

www.quantumuniversity.edu.in

Syllabus

Course Type: Engineering **Course :** **Bachelor of Technology**

Year/sem: **4 Sem** **Branch :** **CSE** **Created By:**

Object Oriented Programming and Systems with Java

CS34212

Unit	Sub Unit	Topic	Duration
1(Object-Oriented thinking)		, messages and methods, Responsibilities, Classes and Instances, Class Hierarchies-, Data types, Variables and Arrays, operators, expressions, control statements,, Introducing classes, Methods and Classes, String handling., A way of viewing world – Agents and Communities, Inheritance, Method binding, Overriding and Exceptions, An Overview of Java,	8
GUI Programming with Swing		Introduction, limitations of AWT, Understanding Layout Managers, Flow Layout, Border Layout, Grid Layout, Card Layout, Grid Bag Layout, MVC architecture, components, containers	7
Inheritance		Inheritance concept, Inheritance basics, Member access, Polymorphism-ad hoc polymorphism, Constructors, Creating Multilevel hierarchy, super uses, using final with inheritance, pure polymorphism, method overriding, abstract classes.	7
Packages		Interfaces- defining an interface, Access protection, importing packages, Defining a Package, CLASSPATH, implementing interfaces, Nested interfaces, applying interfaces, variables in interfaces and extending interfaces, Exception Handling, Multithreading., The Collections Framework (java.util) - Collections overview, Collection Interfaces, The Collection classes.	7
Stream based I/O (java.io)		The Stream classes-Byte streams and Character streams, Reading console Input and Writing Console Output, File class, Reading and writing Files,, The Console class, Serialization, Enumerations, auto boxing,, Random access file operations, generics.	7

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